

Can Machine Learning Predict the Stock Market?

An Honest Test of Next-Day to Next-Year Price Forecasting on the S&P 500, NVIDIA, and JPMorgan Chase

Quan
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Abstract

This project asks whether standard machine learning models can predict stock prices better than the simplest possible guess: that tomorrow's price will equal today's. Using ten years of daily price data from Yahoo Finance for the S&P 500 index, NVIDIA, and JPMorgan Chase, twelve features were computed from past prices and volume (recent returns, volatility, moving-average ratios, RSI, and a volume z-score), and two models, ridge regression and a random forest, were trained to predict the logarithmic return over five horizons: one day, one week, one month, one quarter, and one year. Models were evaluated strictly in time order, with the most recent 250 trading days held out and a purge gap to prevent information leaking from the future. In a walk-forward backtest of the most recent trading year, the best model beat the naive baseline only by a small margin (for example, a test-set RMSE of 0.00770 versus 0.00778 for the S&P 500 at the one-day horizon, with 56.4% direction accuracy), and for JPMorgan Chase the naive baseline won outright. A one-year hindcast made with only data available on June 10, 2025 predicted the S&P 500 would rise 17.8%; it actually rose about 22.4%, landing inside the model's stated uncertainty band. The results are consistent with the Efficient Market Hypothesis: short-horizon price movements are dominated by noise, and apparent long-horizon skill mostly reflects the market's general upward drift. The project's main contribution is methodological: it demonstrates how easily forecasting projects can fool their authors, and how careful evaluation design prevents it.

1. Introduction

Predicting the stock market is one of the most tempting applications of machine learning, and also one of the easiest to get wrong. A model that appears to predict prices with high accuracy is usually not a breakthrough; it is usually a bug. This paper documents Veritas, a small forecasting system built to test, as honestly as possible, whether two standard machine learning models can out-predict a baseline so simple it sounds like a joke: tomorrow's price equals today's price.

Three assets were chosen to represent different levels of difficulty: the S&P 500 index (ticker ^GSPC), a broad basket of 500 large American companies; NVIDIA (NVDA), a famously volatile technology stock; and JPMorgan Chase (JPM), a large bank. For each asset, separate models were trained to predict the total return over the next trading day, week (5 trading days), month (21), quarter (63), and year (252).

The central hypothesis, motivated by the Efficient Market Hypothesis (Fama, 1970), was that the models would beat the naive baseline only slightly, if at all, at short horizons, and that any apparent skill at long horizons would mostly be explained by the market's tendency to drift upward over time. As the results show, both predictions held.

2. Background

2.1 The Efficient Market Hypothesis

The Efficient Market Hypothesis (EMH) states that asset prices already reflect available information. If tomorrow's price rise were predictable from today's public data, traders would buy today, pushing the price up immediately and erasing the predictable part. The consequence is that day-to-day price changes should look mostly like random noise (Fama, 1970; Malkiel, 2003). The EMH does not forbid prediction entirely, but it predicts that any edge found in past prices alone will be tiny. This makes it the natural null hypothesis for this project.

2.2 Why predict returns instead of prices

A raw price chart drifts upward for years, so a model trained on prices mostly learns that the number goes up, which is not a useful skill. Veritas instead predicts the logarithmic return, $\log(\text{future price} / \text{today's price})$, which is approximately the percent change and looks statistically similar in 2016 and 2026. Converting a predicted log return r back into a price is one line: $\text{predicted price} = \text{today's close} \times e^r$.

2.3 The models

Two scikit-learn models were used (Pedregosa et al., 2011). Ridge regression is linear regression with a penalty ($\alpha = 10$) that shrinks coefficients toward zero, acting as a safety brake against overfitting (Hoerl & Kennard, 1970). The random forest is an ensemble of 300 decision trees that vote together (Breiman, 2001); the trees were deliberately kept shallow (maximum depth 4, at least 20 samples per leaf) because deep trees memorize noise. Both models sit in a pipeline that first standardizes every feature to mean 0 and standard deviation 1, so no feature dominates just because its numbers are bigger. They compete against the naive baseline, which always predicts a return of exactly zero. That baseline sounds unbeatable only until one remembers what the EMH implies: for daily prices it is famously hard to beat.

3. Data and Methods

3.1 Data

Ten years of daily open, high, low, close, and volume history per ticker were downloaded from Yahoo Finance via the yfinance library, adjusted for splits and dividends. This is roughly 2,500 trading days per asset. All experiments reported here use data through June 11, 2026.

3.2 Features

For each trading day, twelve features were computed using only information available on or before that day's close (Table 1). Letting the model accidentally see the future, called lookahead bias or data leakage, is the single most common way such projects go wrong, so the code includes an automated test that doubles all future prices and verifies that no feature for earlier days changes.

Feature	What it measures
Returns over last 1, 2, 3, 5, 10 days	Recent momentum at several time scales (5 features)
Volatility, 5- and 21-day	Standard deviation of recent daily returns: how shaky the price has been (2 features)
Price vs. 10- and 50-day average	Percent above or below the moving average (2 features)
RSI (14-day)	Relative Strength Index: classic overbought/oversold momentum indicator, scaled 0 to 1
Volume z-score (21-day)	Whether today's trading volume is unusually high or low
Day of week	Monday = 0 to Friday = 1; markets sometimes behave differently by weekday

Table 1. The twelve input features, all computed from past data only.

3.3 Honest evaluation design

The data were split by time, never shuffled: the most recent 250 trading days (about one year) were held out for testing, and everything earlier was used for training. Shuffling rows, as in a typical machine learning tutorial, would let the model train on days that come after its test days, which inflates scores and makes them meaningless. For horizons longer than one day, a purge gap equal to the horizon was inserted between the training and test sets, because the last training rows' targets reach into the test period and would otherwise give the model a sneak peek.

Beyond the single split, a walk-forward backtest replayed the final year one day at a time: models were refit every 21 trading days using only data available up to that morning, predicted the next day, and were graded before moving on. A second leakage test doubles all future prices and confirms earlier walk-forward predictions do not budge. Finally, a hindcast rewound the entire system to June 10, 2025, regenerated the full five-horizon forecast using only data available then, and graded it against what actually happened over the following year.

3.4 Metrics

Predictions were scored with mean squared error (MSE), its square root RMSE (in return units, so 0.0077 means a typical miss of about 0.77% per day), mean absolute error (MAE), the coefficient of determination R-squared (1 is perfect, 0 means no better than guessing the average, negative means worse), direction accuracy (how often the up-or-down call was right; a coin flip scores 50%), and the average dollar error of the implied price prediction. The backtest also simulated a toy strategy: start with \$10,000 and hold the asset only on days the model predicts a gain, compared against simply buying and holding. The simulation ignores trading fees and taxes, which would only make the strategy look worse.

4. Results

4.1 Next-day prediction barely beats the naive baseline

Table 2 shows the held-out test-year scores for the S&P 500 at the one-day horizon, as documented in the project's reference run. The random forest beat the naive baseline on every error metric, but the margin is paper thin: RMSE improved from 0.00778 to 0.00770, about one percent, and the average next-day price error improved from \$39.50 to \$39.25 in index points. R-squared of 0.012 means the model explains roughly 1% of the day-to-day variation. The one number of real interest is direction accuracy: 56.4%, where a coin flip scores 50%. Results for NVIDIA were similar in character, and for JPMorgan Chase the naive baseline produced a lower RMSE than either trained model, meaning that for that stock the best available prediction was no prediction at all.

Model (S&P 500, 1-day)	RMSE	MAE	R ²	Direction
Naive (tomorrow = today)	0.00778	0.00585	-0.009	n/a
Random forest	0.00770	0.00581	0.012	56.4%

Table 2. Held-out test-year scores, S&P 500, one-day horizon (RMSE and MAE in log-return units).

4.2 Walk-forward backtest

Figure 1 shows the walk-forward backtest of the S&P 500 over the final trading year (June 2025 to June 2026). The top panel compares the toy strategy's equity curves with buy and hold: \$10,000 left in the index grew to roughly \$12,000, while following the random forest's daily up/down calls ended near \$11,700 and following ridge regression near \$11,100. Neither model strategy beat simply holding the index, even before subtracting trading fees. The bottom panel shows 50-day rolling direction accuracy: the random forest spent much of late 2025 meaningfully above the 50% coin-flip line, peaking near 65%, then fell below it during the spring 2026 drawdown, a reminder that whatever weak pattern the model found was not stable. For NVIDIA, the model-following strategies roughly tracked buy and hold (all curves ended in the \$14,000-15,500 region), largely because the models predicted gains on most days, which for a rising stock is nearly the same thing as holding it.

One trap matters when reading direction accuracy: markets drift upward, so a strategy of always guessing up already scores well above 50% in a good year. The backtest prints the share of up days precisely so the model can be compared against that smarter benchmark rather than against a coin flip.

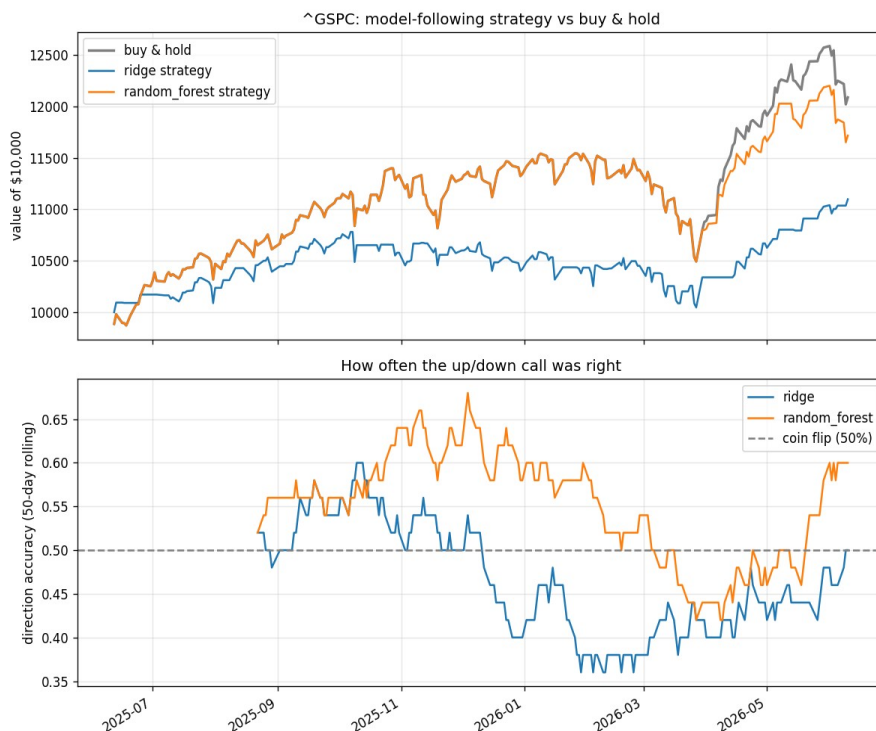


Figure 1. Walk-forward backtest, S&P 500, June 2025 to June 2026. Top: \$10,000 following each model vs. buy and hold. Bottom: 50-day rolling direction accuracy vs. the 50% coin-flip line.

4.3 The five-horizon forecast

Table 3 shows the live forecast generated on June 11, 2026, the most recent trading day in the dataset. Two details deserve attention. First, several short-horizon cells show a change of exactly 0.0%: those are horizons where the naive baseline won the model competition on the held-out year, so the honest forecast is no change. Second, the long-horizon predictions look impressively confident (+16.3% for the S&P 500 in a year, +66.6% for NVIDIA) but carry enormous uncertainty bands, visible as the widening shaded region in Figure 2; the one-year band for NVIDIA spans hundreds of dollars.

Horizon	S&P 500 (7,394.30)	NVIDIA (\$204.87)	JPMorgan (\$313.49)
Tomorrow	7,396.69 (+0.0%)	204.98 (+0.1%)	313.49 (+0.0%)
Next week	7,413.76 (+0.3%)	204.87 (+0.0%)	314.98 (+0.5%)
Next month	7,437.41 (+0.6%)	215.02 (+5.0%)	315.45 (+0.6%)
Next quarter	7,686.55 (+4.0%)	233.52 (+14.0%)	324.45 (+3.5%)
Next year	8,597.46 (+16.3%)	341.41 (+66.6%)	366.17 (+16.8%)

Table 3. Predicted closing prices made on June 11, 2026 (closing price that day in parentheses). Cells with +0.0% are horizons where the naive baseline won.

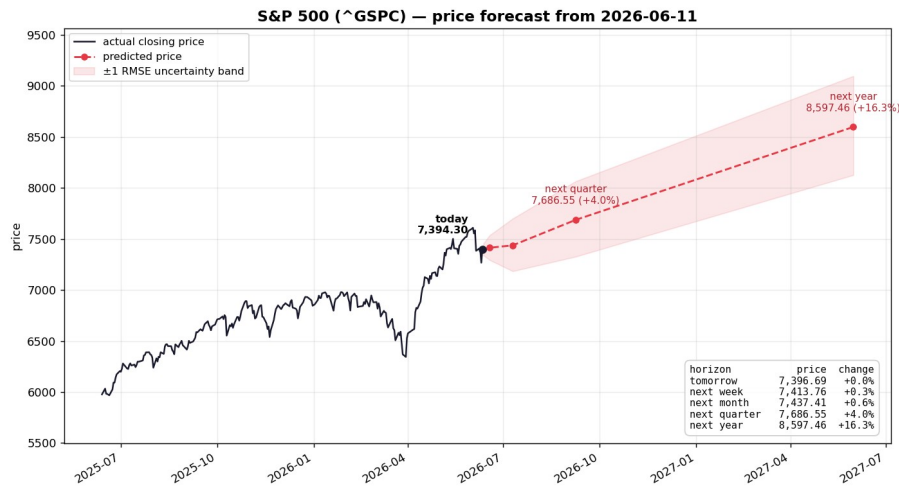


Figure 2. S&P 500 forecast from June 11, 2026, with the ± 1 RMSE uncertainty band widening at longer horizons.

4.4 Hindcast: grading a year-old forecast

The strongest test in the project is the hindcast: the entire system was rewound to June 10, 2025 and asked to forecast the following year using only data available then. Table 4 grades the one-year predictions against what actually happened. The S&P 500 forecast of +17.8% versus an actual +22.4% is a genuine, if partial, success: right direction, plausible magnitude, and the realized path stayed inside the forecast's ± 1 RMSE band (Figure 3). NVIDIA shows the opposite failure mode: the model predicted +66.8% and the stock gained 42.5%, a 24-percentage-point overshoot, and at the week, month, and quarter horizons the winning model was the naive baseline, so those forecasts were flat. JPMorgan was undershot (+11.1% predicted versus +19.0% actual). All three calls got the direction right, but as Section 5 argues, that is faint praise.

Ticker	Price 2025-06-10	Predicted 1 yr later	Actual 2026-06-11
S&P 500	6,038.81	7,116.64 (+17.8%)	7,394.30 (+22.4%)
NVIDIA	143.76	239.75 (+66.8%)	204.87 (+42.5%)
JPMorgan	263.48	292.79 (+11.1%)	313.49 (+19.0%)

Table 4. One-year hindcast made on June 10, 2025, graded against reality.

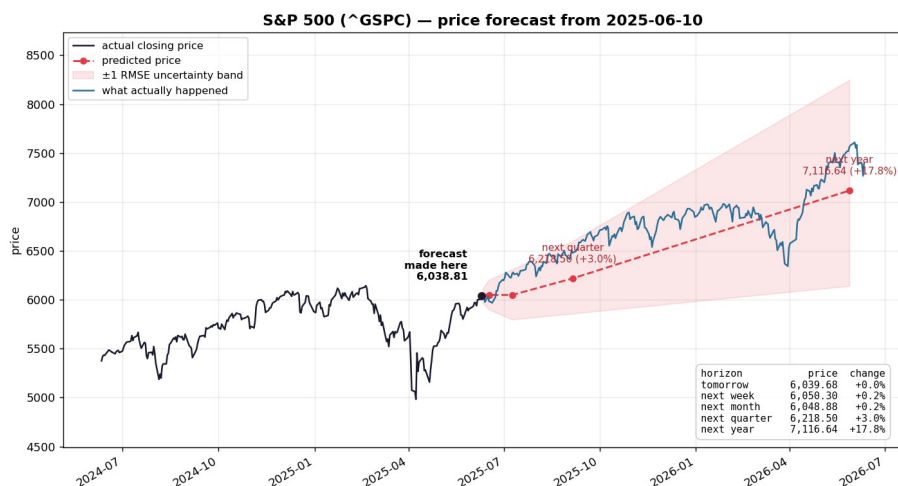


Figure 3. The June 2025 hindcast (red dashed) vs. what actually happened (blue). The realized path stays inside the ± 1 RMSE band.

5. Discussion

The headline result is a negative one, and that is the point. At the one-day horizon, ten years of data, twelve sensible features, and two competent models bought an RMSE improvement of about 1% over predicting nothing, and for one of the three assets they bought less than nothing. This is exactly the world the Efficient Market Hypothesis describes: information in past prices is almost fully absorbed into current prices, leaving behind noise with at most a faint signal. The 56.4% direction accuracy on the S&P 500 is interesting but should be read alongside two caveats: an always-guess-up strategy scores roughly 55% in a typical rising year, and the toy trading strategy built on those calls still failed to beat buy and hold even with zero trading fees.

The long-horizon results illustrate a subtler trap. A one-year model that predicts +16% to +18% for a broad index looks skillful, and the hindcast even landed within its error band. But stocks drift upward on average, so a model can score well at the one-year horizon by learning little more than markets usually go up. High direction accuracy at long horizons proves the drift exists, not that the model is clairvoyant. The honest way to report such forecasts is with their uncertainty: the printed RMSE-based band for the one-year S&P 500 forecast spans roughly 1,000 index points, which is the difference between a dull year and a great one.

Three design decisions did the most work in keeping the results trustworthy. First, predicting log returns rather than prices removed the upward drift that lets models fake skill. Second, splitting by time with a purge gap, and backtesting walk-forward, ensured no prediction ever used future information; both properties are enforced by automated tests rather than by hope. Third, requiring every model to beat the naive baseline, and letting the baseline win, prevented the strongest temptation in forecasting projects: declaring victory because a model produces numbers. A suspiciously good result, such as a daily-return R-squared near 0.3, would have indicated a data leak rather than a discovery.

Limitations remain. One test year is a small sample, and a single good or bad year can dominate the backtest. The toy strategy ignores fees, taxes, and slippage, all of which penalize frequent trading. The feature set uses only price and volume history; public information such as earnings, news, or interest rates might carry signal the models never saw. And overlapping windows at long horizons make those error metrics less independent than they appear. These are reasons to distrust the optimistic parts of the results even further; none of them rescue the pessimistic parts. This project is a learning exercise, and nothing in it should be used to trade real money.

6. Conclusion

Can machine learning predict the stock market? On the evidence here: barely, sometimes, and not profitably. Standard models given ten years of price history edged out the naive baseline by about 1% at the one-day horizon on two of three assets, lost to it outright on the third, and failed to beat buy and hold in a fee-free simulated year. A year-old forecast of the S&P 500 landed within its own error band, but most of that success is the market's upward drift wearing a model's clothing. The project's real product is not a prediction but a method: time-ordered evaluation, purge gaps, leakage tests, naive baselines, and uncertainty bands. Those tools are what separate an honest negative result from a flattering wrong one, and they transfer to any forecasting problem a student will meet.

References

- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5-32.
- Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2), 383-417.
- Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 55-67.
- Malkiel, B. G. (2003). The efficient market hypothesis and its critics. *Journal of Economic Perspectives*, 17(1), 59-82.
- Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
- Yahoo Finance. (2026). Historical daily price data for ^GSPC, NVDA, and JPM, accessed June 11, 2026 via the yfinance library. <https://finance.yahoo.com>